Joe Ledford
Associate Software Engineer
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PROFESSIONAL SUMMARY

Associate Software Engineer with experience building machine learning pipelines and intelligent software systems that combine AI, data engineering, and enterprise software development. Currently expanding expertise in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), with a passion for delivering intelligent enterprise systems that align technical innovation with business strategy.

SKILLS

Technical Stack

- Programming Languages: Python, SQL, R, Stata
- AI/ML: Scikit-learn, LangChain, PyTorch, Hugging Face
- Data Engineering: PySpark, SQLAlchemy, FastAPI, MySQL, DBeaver
- Version Control: Git

Engineering

- SDLC
- Testing & Debugging
- Technical Documentation
- Enterprise AI Systems
- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)

PROFESSIONAL EXPERIENCE

Tata Consultancy Services — **Remote**
Associate Software Engineer — *June 2026 – Current*

- Collaborated with cross-functional teams to translate business requirements into technical designs, implement features, and ensure successful testing throughout SDLC.
- Develop Python-based software components supporting enterprise manufacturing applications using enterprise software engineering best practices.
- Contribute to the development and evaluation of AI-powered solutions utilizing Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) architectures for enterprise knowledge retrieval and intelligent automation.
- Assist with the design and integration of data ingestion, retrieval, and prompt orchestration pipelines supporting LLM-based applications.

IDX Exchange — **Remote**
Data Science Intern — *January 2026 – May 2026*

- Built an end-to-end machine learning pipeline in Python for residential home price prediction using structured real estate transaction data from the Cotality (Trestle) API.
- Executed data ingestion, cleaning, feature engineering, model training, and evaluation workflows with pandas and scikit-learn, enhancing data quality for analysis.
- Trained and compared multiple regression models (Linear, Random Forest, CatBoost), focusing on optimizing R^2 and MAPE to improve predictive accuracy.
- Applied feature engineering and preprocessing techniques including missing value handling, encoding strategies, and transformation of skewed variables.
- Presented model performance updates and results in weekly technical team meetings, integrating feedback to refine models and enhance outcomes.

Tatte — **Bethesda, MD**
Dishwasher/FOH — *September 2026 – December 2026*

- Coordinated kitchen and front-of-house operations to sustain uninterrupted service during peak hours.
- Operated and maintained commercial dishwashing systems to uphold stringent sanitation and safety standards.
- Maintained efficient flow of clean dishware and kitchen tools in a high-volume service environment.

EDUCATION

American University, College of Arts and Sciences — **Washington, DC**
Bachelor of Science: Data Science, Economics Minor | GPA: 3.8 — *Aug 2023 – May 2026*

- Relevant Coursework: Statistical Machine Learning, Applied Econometrics II, Computer Science II, Applied NLP

University of California, Santa Cruz — **Santa Cruz, CA**

Philosophy Major| GPA: 4.0 *Sep 2022 –April 2023*

- Relevant Coursework: Philosophy of Cognitive Science, Environmental Ethics, Classical Mythology

TECHNICAL PROJECTS

Kairos — Statistical Arbitrage Trading System *(Ongoing personal project)*

Python | pandas | statsmodels | Alpaca API | LangGraph | FastAPI | MySQL | Azure

- Built a pairs-trading research pipeline that screens 150+ ticker combinations via Engle-Granger cointegration testing (ADF, OLS regression) to identify statistically mean-reverting equity pairs, filtered against fundamental economic rationale rather than statistics alone.
- Designed a rolling z-score signal generator and backtesting engine computing Sharpe ratio, maximum drawdown, and cumulative P&L, using real historical market data pulled via the Alpaca API.
- Implementing a Kalman filter to replace the static OLS hedge ratio with a time-varying estimate, benchmarked directly against the static baseline.
- Building out MySQL persistence for pair statistics, trade decisions, and backtest results
- Architecting an agentic risk-overlay layer (LangGraph) that screens news and SEC filings for fundamental breaks in a pair's statistical relationship before trade execution — with a planned FastAPI service layer and Azure containerized deployment.

Housing Valuation Pipeline

Python | CatBoost | scikit-learn

- Engineered a reusable end-to-end machine learning pipeline for forward-tested residential property valuation across **166,000+** California real estate transactions, enabling robust model evaluation on future datasets.
- Designed modular data preprocessing and feature engineering workflows incorporating haversine distance calculations, target encoding, logarithmic transformations, and statistically driven imputation strategies to improve predictive performance and generalization.
- Developed a comprehensive model validation framework leveraging forward testing, multicollinearity analysis, correlation auditing, and performance benchmarking, achieving **89.6% R^2**, **11.3% MAPE**, and **7.6% MdAPE** using CatBoost.

Content-Based Music Recommendation System

Python | NLP | Spotify API | Genius API | LyricsBERT

- Developed a content-based music recommendation system by integrating the Spotify and Genius APIs to automatically construct a personalized lyrics dataset from user listening history.
- Evaluated progressively more sophisticated natural language processing approaches—including TF-IDF, Latent Semantic Analysis (LSA), and LyricsBERT embeddings—to compare recommendation quality across multiple model architectures.
- Designed a modular recommendation framework that highlighted the trade-offs between traditional vector-space methods and transformer-based language models while supporting future experimentation with additional NLP techniques.

Time Series Forecasting of Peak Cherry Blossom Bloom

Python | Time Series | Econometrics

- Designed an interpretable phenology forecasting framework for the **Washington Statistical Association/George Mason University Cherry Blossom Prediction Competition**, iteratively refining models from Quetelet and Growing Degree Day approaches to a Parallel Threshold Dormancy model.
- Identified and corrected data leakage in early model iterations while incorporating contemporary phenology research to model the dynamic interaction between winter chilling and spring heat accumulation under changing climate conditions.
- Applied rolling-origin cross-validation and systematic model evaluation to improve forecast robustness, reduce overfitting, and support scientifically grounded prediction of peak bloom dates.